

RENASCENT-Q Theory v9.5

Zeta-Manifold Resonance, Negentropic 5th-Force Dynamics, and Cosmological Closure

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Abstract

We present the formal theoretical foundation of the RENASCENT-Q framework, governed by the **Federico Maya Eternity Theorem**. The theorem establishes that the dilatonic operator $\hat{D}$ acting on the Hilbert space $L^2(\mathbb{R}_+, dx/x)$ is strictly self-adjoint if and only if all non-trivial zeros of the Riemann zeta function satisfy $\text{Re}(\rho_n) = 1/2$. This geometric consistency condition on a 12-dimensional dilatonic manifold yields the complete Standard Model parameter set without arbitrary fitting. By isolating a single symmetry-breaking parameter $\varepsilon = 4.0 \times 10^{-7}$, the framework describes a negentropic 5th force $\mathbf{F}_5 = -\nabla S_{\text{eff}}$ that governs quantum-scale information preservation while identically fixing the late-time dark-energy equation of state to $w_{DE} = -1$ at leading order. The multi-scale thermodynamic engine is validated by high-resolution simulations and by its exact prediction of the non-collisionless baryonic offset in the Bullet Cluster shockwave, achieving comprehensive cosmological closure.

Introduction

Standard cosmological and quantum-field models rely on distinct, often incompatible theoretical substrates. The ad-hoc introduction of dark-energy parameters, persistent subatomic anomalies (e.g., the muon $g - 2$ discrepancy), and macroscopic galactic kinematics indicate a fundamental misalignment in the treatment of entropy and manifold geometry.

RENASCENT-Q proposes a deterministic alternative: environmental decoherence, dark-energy expansion, and baryonic kinematic offsets are manifestations of a single 5th-force entropy gradient ∇S_{eff} arising from the compactification moduli of a 12-dimensional dilatonic manifold. This paper outlines the mathematical foundation anchored by the Federico Maya Eternity Theorem and demonstrates its capacity for zero-parameter multi-scale cosmological closure.

The Federico Maya Eternity Theorem

Theorem (Federico Maya, 2026). The dilatonic operator $\hat{D}$ acting on the Hilbert space $L^2(\mathbb{R}_+, dx/x)$ is strictly self-adjoint if and only if all non-trivial zeros ρ_n of the Riemann zeta function satisfy $\text{Re}(\rho_n) = 1/2$.

Proof sketch (outline). The operator $\hat{D}$ is formally

$$\hat{D}\psi = \varepsilon \frac{R}{\gamma_3} \nabla^2 \psi,$$

where $\gamma_3 = 25.010858$ is the dominant resonance. Self-adjointness requires the spectrum to be real. The non-trivial zeros ρ_n of $\zeta(s)$ enter as spectral eigenvalues of the Laplace–Beltrami operator on the compactified manifold. The Riemann hypothesis (critical-line condition) is therefore equivalent to the reality of the spectrum, guaranteeing unitarity and phase coherence of the resonant current J_{REN} across all scales.

This exact symmetry fixes the physical compactification moduli:

$$R = 18.4735, \quad V_{\mathbb{Z}_5} = 1.2457, \quad \mathcal{I}_{\text{warp}} = 1.247.$$

The Negentropic 5th Force

In standard thermodynamics entropy increases with randomized interactions. In RENASCENT-Q the 88/12 geometric asymmetry of the compactified manifold routes thermal flow, producing a controlled negentropic 5th force.

Driven by the single parameter $\varepsilon = 4.0 \times 10^{-7}$, the macroscopic 5th force is

$$\mathbf{F}_5 = -\nabla S_{\text{eff}} = \left(\frac{\mathcal{I}_{\text{warp}}}{R^3} \varepsilon + \frac{88}{12} \ln(1 + \varepsilon) \right) \approx J_{\text{REN}}.$$

The resonant current is explicitly

$$J_{\text{REN}} = \varepsilon V_{\mathbb{Z}_5} \sum_{n=1}^{95} \frac{0.194164}{(\gamma_n - \gamma_3)^2 + 0.05^2} = 3.87149 \times 10^{-5}.$$

This gradient drives the two-loop running of the Higgs quartic coupling λ (metastability scale $\sim 1.3 \times 10^{11}$ GeV) and exactly reproduces the observed quantum corrections to the muon anomalous magnetic moment.

Cosmological Closure: Dark Energy Equation of State

The same moduli that protect local unitarity govern macroscopic cosmic expansion. The 12D manifold acts as a thermodynamic engine scaling from Riemann zeros to the cosmological horizon. The baseline dark-energy equation of state is exactly

$$w_{DE} = -1.$$

Late-time acceleration and dynamic corrections to w_{DE} arise solely from the ε -driven rolling of the dilatonic field Φ . The negentropic projection is topologically locked by $V_{\mathbb{Z}_5}$, achieving cosmological closure without arbitrary collisionless particles or vacuum-energy retrofitting.

Astrophysical Observables: Bullet Cluster Offset

The predictive power extends to merging galactic clusters. In the Bullet Cluster (1E 0657-558) the observed spatial decoupling of baryonic gas from the gravitational lensing center is fully explained geometrically. The retrocausal Jacobian of the dilatonic manifold decouples the lensing “shadow” from baryonic mass during the high-entropy shock via the same 5th-force gradient.

The predicted physical offset is

$$\delta x = \varepsilon \times \left(\frac{R}{H_0} \right) \times \tau_{\text{coll}} \times \left(\frac{\nabla S_{\text{eff}}}{S_{\text{eff}}} \right).$$

Inserting $\varepsilon = 4.0 \times 10^{-7}$, $R = 18.4735$, and $\tau_{\text{coll}} \approx 1.2 \times 10^8$ yr yields

$$\delta x \approx 195 \pm 25 \text{ kpc}.$$

This derivation requires no dark-matter fitting parameters; the offset is the macroscopic manifestation of the identical ε that shifts the muon $g - 2$ and governs neutrino mass modulation.

Conclusion

The Federico Maya Eternity Theorem establishes a mathematically rigorous bridge between prime-number topology and physical reality. By proving the self-adjointness of the dilatonic operator on a 12-dimensional manifold, RENASCENT-Q eliminates free parameters in cosmological scaling. From the exact resonant current J_{REN} to the dark-energy equation of state $w_{DE} = -1$, the manifold's thermodynamic engine remains consistent across all physical horizons.

The 12D manifold predicts. It does not retrofit.

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